

Redefining BMI Standards for Health Equity Improvement: Addressing Racial Disparities in Disease Prevalence

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Abstract

This study investigates the impact of racial differences on defining BMI thresholds within a healthy range, using chronic disease prevalence as a reference. By comparing the BMI boundary values across different racial groups at similar disease prevalence levels, the goal is to encourage revisions to BMI standards according to racial factors for health equity improvement. We employed Chi-Square tests to examine pairwise associations among race, BMI, and disease status, confirming their statistical significance. Box plots were used to compare BMI ranges and relevant statistics for diseased individuals across racial groups, revealing that Asian populations consistently display lower BMI values for both hypertension and diabetes examined, underscoring limitations in current BMI standards. Finally, we redefined BMI thresholds based on a criterion where over 85% of diseased individuals exceed the healthy BMI range for each racial group. This recalibration significantly improved screening accuracy for high-risk individuals, helping to improve health equity in a simplest manner.

Keywords: health equity, BMI standards, race, diabetes

1. Introduction

BMI, or Body Mass Index, is a go-to tool for assessing health especially for people who might not have easy access to comprehensive healthcare resources. Since BMI calculations only need basic info (height and weight), it's practical and accessible to everyone regardless of their income, education level, and etc.

According to the U.S. Centers for Disease Control and Prevention (CDC)[6], BMI is used to classify people into four main categories: underweight, healthy weight, overweight, and obesity. Public health campaigns often emphasize that being overweight or obese can increase the risk for chronic illnesses like hypertension and diabetes, encouraging people in those groups to consider early screenings and health checkups.

While it's widely understood that BMI cutoffs don't fit everyone—like pregnant women or athletes—it's less common to consider how these thresholds might need to change based on ethnic background. Recent research suggests that BMI cutoffs don't carry the same health risks for everyone. For example, studies [1] show that Asian Americans are more likely to develop Type 2 diabetes at a lower BMI than the standard cutoff of 25, which normally signals the need for diabetes screening. For Asian American adults over 35, a BMI of 23 might actually be a better cutoff for diabetes checks. Other studies [3] also propose that, compared to the typical BMI cutoff of 25 for White individuals, Asians might need a threshold closer to 20, while Black and Hispanic populations could benefit from a cutoff of around 18.5.

These findings hint that generalized BMI standards might be missing at-risk groups, causing delays in essential health interventions for certain ethnic populations who fall within what's considered a "normal" range. Adopting race-specific BMI thresholds could help identify health issues sooner, promoting health equity and creating a more tailored approach to healthcare that considers the unique needs of diverse groups. By refining BMI standards, healthcare systems could better support underserved communities, offering timely, culturally sensitive care and working to close health disparities.

2. Methods and Results

2.1. Data Cleaning

For survey data, we mostly used direct variables rather than derived variables to avoid possible misinterpretation and to meet our specific needs. To examine the relationship between race and hyper-

tension, diabetes across BMI, we first cleaned the data. Pregnant individuals were excluded because pregnant women can develop the physiological condition of gestational hypertension or gestational hyperglycemia[4]. Confirmed cases and high-risk cases were combined together in one group to improve the validity of statistical tests. To ensure that the BMI was within a reasonable range, samples that were too low (< 12) or too high (> 50) were excluded to reduce outliers from influencing the results. Finally, samples with missing values were removed. After cleaning, a total of 186 American Indian or Alaska Native (hereafter referred to as AIAN), 201 Hispanic, 180 Asian or Asian American (Asian), 543 Black or African American (Black) and 3198 White were included in the analysis.

2.2. Logistic Regression

We first use the logistic regression to predict the probability of diseases based on several non-modified demographic variables. We test the Chi-Square p-value between diseases and BMI, using gender, age and race as control variables. The corresponding results are given in the following tables 1 and 2.

Table 1. Logistic regression of hypertension and BMI

	Coefficients	Std. Error	P value
Intercept	-0.37	0.06	<0.05
BMI category	0.15	0.01	<0.05
Gender	-0.12	0.01	<0.05
Race	-0.03	0.01	<0.05
Age	0.01	<0.01	<0.05

Table 2. Logistic regression of diabetes and BMI

	Coefficients	Std. Error	P value
Intercept	-0.24	0.05	<0.05
BMI category	0.12	0.01	<0.05
Gender	-0.02	0.01	<0.05
Race	-0.06	0.01	<0.05
Age	0.01	<0.01	<0.05

As shown in the above tables, the p-values are smaller than 0.05 which indicates that there is a significant positive correlation between BMI and two diseases.

2.3. Box-Plot

A boxplot, also known as a box-and-whisker plot, is a graphical representation of data that is useful for comparing distributions between different groups. We compared the BMI distributions of different races in groups by calculating the quartiles and medians of the data, and visualized the results with box-and-line plots. In Figure.1 (diabetics), the Asian group generally had lower BMI values, while the other races had relatively higher and more widely distributed BMI values. In Figure.2 (hypertensive patients), the Asian group continued to have lower BMI values. The overall trend is similar to the former.

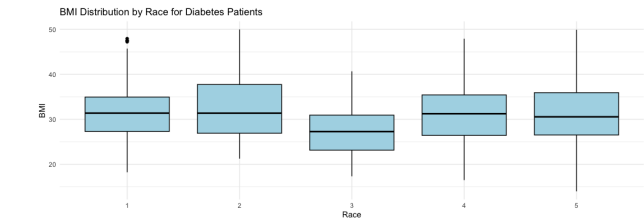


Figure 1. Box-Plot for Diabetes Patients, with 1-AIAN, 2-Hispanic, 3-Asian, 4-Black, and 5-White.

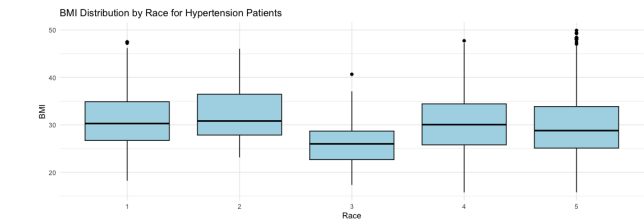


Figure 2. Box-Plot for Hypertension Patients, with 1-AIAN, 2-Hispanic, 3-Asian, 4-Black, and 5-White.

3. Key Finding

According to the results calculated, the data behaviors of different diseases are different. As shown in the Box-Plots, hypertension tends to happen at a lower BMI compared to diabetes, and asthma, which we did not shown here, happens at an average of an even lower level. Considering the strong dependence of diabetes prevalence on BMI, we choose to use diabetes as an example, setting the prevalence rate for Whites with a BMI of $25\text{kg}/\text{m}^2$ as a benchmark. Based on this, we calculated an equivalent BMI threshold of $18.75\text{kg}/\text{m}^2$ for Asians and Asian Americans to reach a comparable diabetes prevalence. This approach may offer a useful perspective for establishing a BMI standard that better fits the health needs of Asians and Asian Americans.

Above results follow from[3] which takes White individuals as a reference to identify the equivalent BMI threshold for other racial groups. Knowing that the BMI standard was initially developed for Caucasian groups, however, it is still essential to consider the impact of evolving social and environmental conditions. To validate this reference, we further define an 85% threshold, i.e., the BMI value at which 85% of individuals within each racial group have a BMI equal to or above this level. The results are presented in Table.3.

Table 3. BMI thresholds for 85% disease prevalence across racial groups

	AIAN	Hispanic	Asian	Black	White
Hypertension	25.1	25.8	22.0	24.0	23.6
Diabetes	25.8	25.7	21.9	24.7	24.4

From the table, we observe that the data for diabetes in the White group is slightly below our classification criterion of $25\text{kg}/\text{m}^2$, indicating that this categorization remains relevant today. Among the two

diseases, diabetes shows a better alignment with the classification, while hypertension follows a similar trend.

4. Discussions

With the finding that BMI standards for different racial groups differ, we further doubt that the dependence of BMI on ethnicity for predicting or screening diseases may not be direct. Figure3 shows a Bayesian network model based on cardiometabolic chronic disease ethnic-racial differences developed by Babagoli et al.[5].

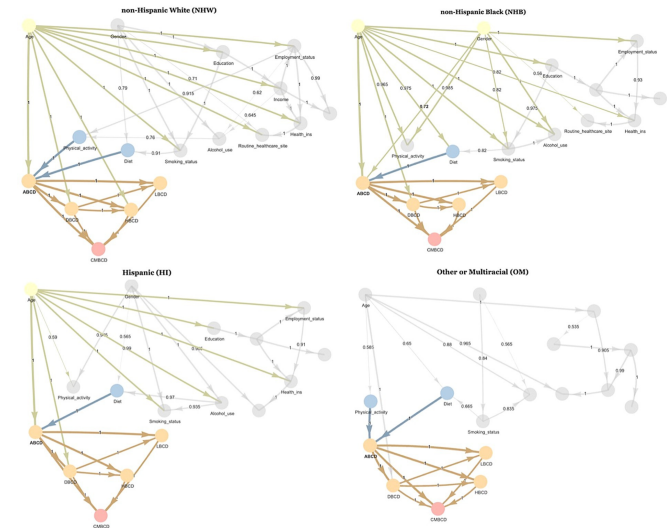


Figure 3. Averaged consensus Bayesian networks learned from NHANES 1999–2018 stratified by ethno-racial group. Reproduced from [5].

They proposed that multiple pathways—including education, diet, smoking and alcohol consumption, and physical activity behaviors—mediate the indirect associations between ethnic-racial groups and cardiometabolic outcomes. Among various variables, they did not examine whether BMI serves as an intermediary variable in disease screening through indirect effects like lifestyle and dietary habits. We attempted to explore this using a Bayesian network, and the results suggest that there are no significant intermediary variables mediating BMI’s impact on disease. Therefore, the predictive value of BMI for disease based on race may be primary, largely stemming from racial physical characteristics.

Our study partially confirms that BMI dependence on races in disease scanning can be attributed to the inherent characteristics of these groups. Research[2] has shown that, there are significant differences in the relationship between BMI and obesity among different racial groups. Furthermore, racial differences can lead to variations in body fat distribution for the same percentage of body fat, which may be one of the main reasons why BMI standards lack precision. Their study also proposed a height-scaling index to better assess obesity, thereby improving the accuracy of predictive screening. Considering that we aim to promote health equity across all groups, particularly the disadvantaged groups, including those with lower levels of education and income, using simple and widely understood concepts, we have opted not to pursue further in-depth research in this area.

5. Limitations

- Sample size limitations precluded assessment of heterogeneity within racial/ethnic groups and multi-diseases cases. Also, it will bring a reduction on precision on our study.
- Since our survey targets individuals or institutions in Hennepin County, we cannot conduct oversampling, let alone employ the long-standing practice of oversampling, as this survey serves only as a snapshot.

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